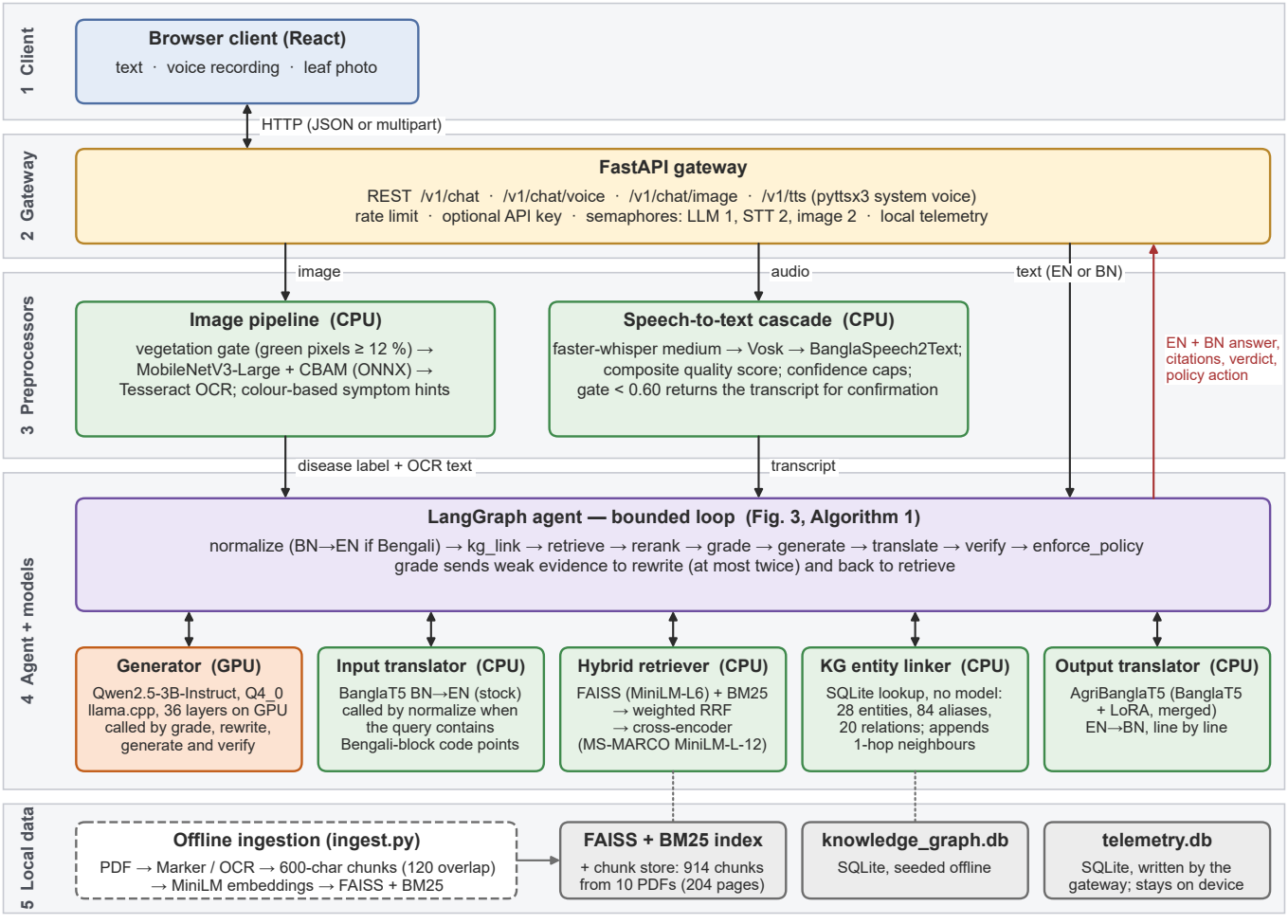

1 Client
Browser client (React)
text · voice recording · leaf photo
HTTP (JSON or multipart)
2 Gateway
FastAPI gateway
REST /v1/chat · /v1/chat/voice · /v1/chat/image · /v1/tts (pyttsx3 system voice)
rate limit · optional API key · semaphores: LLM 1, STT 2, image 2 · local telemetry
image
audio
text (EN or BN)
3 Preprocessors
Image pipeline (CPU)
vegetation gate (green pixels ≥ 12 %) →
MobileNetV3-Large + CBAM (ONNX) →
Tesseract OCR; colour-based symptom hints
Speech-to-text cascade (CPU)
faster-whisper medium → Vosk → BanglaSpeech2Text;
composite quality score; confidence caps;
gate < 0.60 returns the transcript for confirmation
EN + BN answer, citations, verdict, policy action
disease label + OCR text
transcript
4 Agent + models
LangGraph agent — bounded loop (Fig. 3, Algorithm 1)
normalize (BN→EN if Bengali) → kg_link → retrieve → rerank → grade → generate → translate → verify → enforce_policy
grade sends weak evidence to rewrite (at most twice) and back to retrieve
Generator (GPU)
Qwen2.5-3B-Instruct, Q4_0
llama.cpp, 36 layers on GPU
called by grade, rewrite, generate and verify
Input translator (CPU)
BanglaT5 BN→EN (stock)
called by normalize when the query contains Bengali-block code points
Hybrid retriever (CPU)
FAISS (MiniLM-L6) + BM25
→ weighted RRF
→ cross-encoder
(MS-MARCO MiniLM-L-12)
KG entity linker (CPU)
SQLite lookup, no model:
28 entities, 84 aliases,
20 relations; appends
1-hop neighbours
Output translator (CPU)
AgriBanglaT5 (BanglaT5
+ LoRA, merged)
EN→BN, line by line
5 Local data
Offline ingestion (ingest.py)
PDF → Marker / OCR → 600-char chunks (120 overlap)
→ MiniLM embeddings → FAISS + BM25
FAISS + BM25 index
+ chunk store: 914 chunks
from 10 PDFs (204 pages)
knowledge_graph.db
SQLite, seeded offline
telemetry.db
SQLite, written by the gateway; stays on device
GPU-resident model
CPU-pinned model or lookup
orchestration
data on disk
offline process
request path
response path
index / database access